

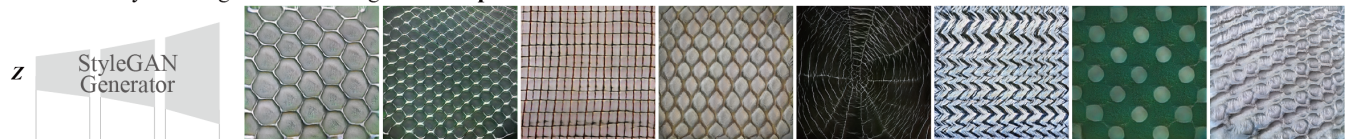
Generation of traditional Japanese patterns from natural patterns with StyleGAN

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1. Pre-train StyleGAN generator on large **natural pattern** dataset



2. Combine the parameters learned on a small **wagara** dataset at the fine level

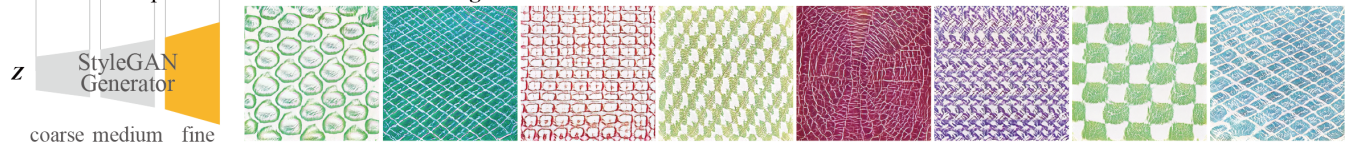


Figure 1: Overview of traditional Japanese pattern (wagara) images generation with StyleGAN. The gray parts of the generator show the parameters learned on natural patterns, and the orange part shows the parameters learned on wagara.

CCS CONCEPTS

• **Computing methodologies** → **Computer vision**; • **Human-centered computing** → **Interaction design**; • **Social and professional topics** → **Professional topics**.

KEYWORDS

Wagara, Pattern, Texture, GAN, Neural Networks

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1 INTRODUCTION

Generative adversarial networks (GANs), which have recently improved the generation of realistic images, heavily depend on a large amount of training data. In various domains, only a limited number of training data are available; thus, the applications of GANs often have not been advanced. Pattern image is one of such domains.

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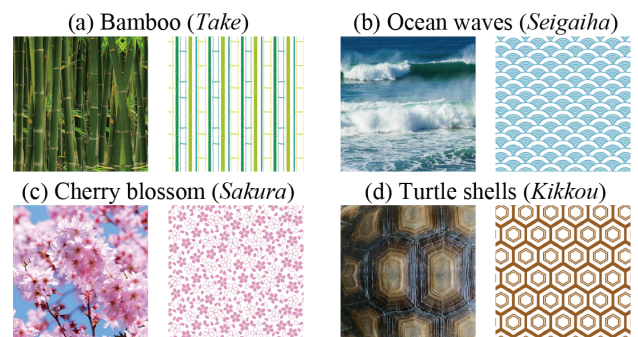


Figure 2: Example wagara images. Most wagara are inspired by patterns and textures in our daily life. In each pair of four examples, wagara (right) is inspired by the scene shown in the left photo. Photos are from Adobe Stock, and wagara images are from [Jogan 2009, 2010].

This study focuses on one type of pattern image, traditional and historical Japanese patterns, often called *wagara*. Wagara is the pattern used for traditional Japanese garments (e.g., kimono and yukata), wallpapers, and other historical items for a long time. Wagara has been created manually, and the creation of wagara requires time-consuming work and professional artistic skills. This paper thus aims to generate wagara automatically with the artificial neural network.

The generation of wagara images with GANs has some difficulties. Firstly, there is no or limited training data. Additionally,

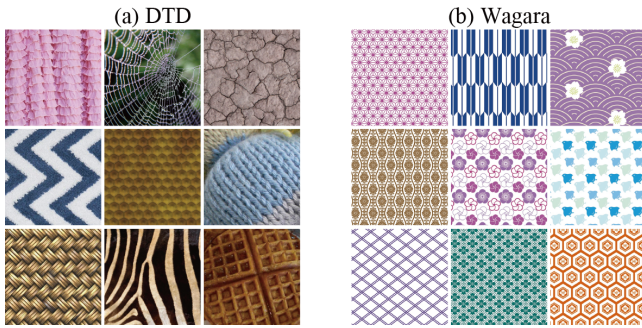


Figure 3: Two datasets used in this work: Describable Textures Dataset (DTD) [Cimpoi et al. 2014] and traditional Japanese patterns (wagara). Wagara dataset is constructed from [Jogan 2009, 2010].

pattern images have characteristics that natural images (e.g., human faces and cars) do not have. Some patterns have self-similarity, and others have regularity.

To overcome these difficulties, we used a large natural patterns dataset as a source dataset. Most wagara images are inspired by patterns and textures in the natural world surrounding our daily lives or created with motifs in the wild such as plants and animals, as shown in Figure 2. For example, *seigaiha* represents ocean waves, and *kikkou* implies turtle shells’ hexagon patterns. Based on this inspiration, we used a dataset containing patterns and textures in the wild as a source dataset. As Figure 1 illustrates the overview, pre-training on the source dataset and transfer learning on a small wagara dataset successfully generated diverse wagara images.

2 DATASETS

We used the Describable Textures Dataset (DTD) [Cimpoi et al. 2014] as a source dataset. This publicly available dataset consists of 5640 pattern and texture images collected in the wild. The example images are shown in Figure 3.

No wagara dataset is available; thus, we made a wagara dataset from two books [Jogan 2009, 2010] and used it as a target dataset. This wagara dataset comprises 258 images containing various colors, shapes, and motifs. All these images are resized to 256×256 . Figure 3 shows the example images of the wagara dataset.

3 METHODS

We used the StyleGAN2 architecture¹ [Karras et al. 2020] and the layer swapping technique [Pinkney and Adler 2020]. Firstly, we trained StyleGAN2 on the source dataset (DTD) with default parameters, and then transfer learning to the target dataset (wagara) created a new StyleGAN2 generator model. Lastly, we split the StyleGAN generator into three levels: coarse, medium, and fine, according to [Karras et al. 2019, 2020]. We combined the parameters and created a new generator following the layer swapping method proposed in [Pinkney and Adler 2020]. At the coarse and medium levels, we used parameters learned on the source dataset; at the fine level, we used parameters learned on the target dataset.

¹<https://github.com/NVlabs/stylegan2>



Figure 4: Curated examples of generated traditional Japanese pattern (wagara) images with our approach.

4 RESULTS

Figure 4 shows the generated wagara images with our approach. Our method successfully generated patterns consisting of repeating structures with wagara-like styles and colors. Figure 1 compares the natural patterns generator and the wagara generator, presenting that traditional Japanese patterns are generated with styles and colors like the constructed wagara dataset. A small amount of training data often causes mode collapse and overfitting; however, in our work, using the large source dataset enables the generation of diverse wagara images.

5 DISCUSSION AND CONCLUSIONS

In this study, we made a dataset of wagara images and generated diverse wagara images with StyleGAN2. Our limitation is the quality: some lines that should be straight are curved, and some areas that should be uniform in color are uneven. Although this work focused on wagara, our approach can be applied to other patterns and textures. Additionally, we plan to create fabrics with the generated designs generated. We hope our work will be helpful for further understanding and application of wagara and other patterns.

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